



Multi-physics modelling of a vapour phase soldering (VPS) system

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HIGHLIGHTS

- Multi-physics model of the Vapour Phase Soldering (VPS) process is presented.
- The model calculates the heat, mass and energy transport mechanisms.
- The model was tested by simulations of a batch type vapour phase soldering system.
- Heating time for the saturation of the vapour concentration and temperature was defined.
- The processing volume of the system was modified according to the simulation.

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ABSTRACT

This paper presents a novel multi-physics modelling approach of the Vapour Phase Soldering (VPS) process. The model applies combined transport mechanisms including heat transport by heat conduction; mass transport by diffusion in the fluids and by phase change during the evaporation and condensation of the Galden liquid; and energy transport caused by mass transport. The model can describe the temperature and vapour concentration relations in the oven – which are the main parameters of the vapour phase soldering process – from the start of the Galden evaporation until the saturation of the vapour space (prepared for soldering). The results calculated by the model show proper conformation with the experimental results of temperature and pressure measurements. The model was tested on a batch type VPS system where the position of the processing volume was optimized in the soldering tank. With the information obtained from the model the whole VPS process can be investigated.

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1. Introduction

Vapour Phase Soldering (VPS, also Condensation Soldering) is an alternative method of conventional reflow soldering techniques, such as forced convection or infrared (IR) type reflow soldering. During the process of VPS, heat transfer fluid is heated at the bottom of a tank, which is the source material of vapour blanket generation. A cooling appliance condenses any excessive vapour at the top of the construction, thus the vapour blanket stays inside the tank. This blanket of generated vapour is the actual heat transfer medium for soldering; therefore the most important parameters are the temperature and the Galden concentration in the vapour space. In this application vapour concentration means the amount of Galden vapour in a given volume. The assembled board is lowered into the vapour, which condenses on the cooler surface of the

board. The latent heat of condensation and the heat transfer of the condensed Galden liquid provide the energy for the solder material to melt. After a specific amount of time, the board is lifted from the vapour to a cooling area.

The VPS technology was invented by Pfahl and Ammann [1] and it was improved later by Wenger who also compared [2] various heat transfer fluids (and their vapours) for the VPS process. However, the VPS technology did not become widespread due to the environmental hazard of Chlorofluorocarbon (CFC) gas by-products, which were released during the process [3]. Vapour phase soldering has come back into focus with the development of “Galden” fluid [4] which is a specific product composed of perfluoropolyether substance (PFPE):



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Nomenclature

T	temperature, K
k	ratio of saturation mass transport
E	energy, J
n	indexing of the mesh
t	time, s
v, w	general variables
R	heat conduction resistance, K/W
α	constant parameter
C	thermal capacity, J/K
C_s	specific heat capacity, J/kg K
pc	phase change
ρ	density, kg/m ³
ev	evaporation
φ	concentration of the vapour, kg/m ³
bp	boiling point

V	volume, m ³
co	condensation
A	surface, m ²
diff	diffusion
m	mass, kg
st	saturation
λ	specific thermal cond., W/m K
txp	thermal expansion
h	latent heat, J/kg
vp	vapour
D	diffusion constant, m ² /s
liq	liquid
β	thermal volumetric coef., m ³ /m ³ K
sol	solid
$\Delta x, \Delta y, \Delta z$	resolution of the mesh, mm
ga	galden

where the flexible ether chain structure is closed with strong carbon-fluorine bonds providing excellent stability for VPS [5]. Galden is considered a harmless, inert material, which is noncorrosive, nontoxic, nonflammable, having zero ozone depletion potential (Zero ODP). Galden is used in many other industrial applications for hermetic sealing [6], heat transfer, semiconductor cleaning and testing [7].

The main advantages of the VPS process are the uniform heating (and heat recovery) of the assembled boards where the maximum temperature is limited [8] because the temperature of the vapour is not able to exceed the boiling point temperature of the fluid itself. Further advantages are the inert atmosphere (fluxes with lower activation level can be used [9]) and the presence of a condensed film layer which keeps the oxygen and other gases from coming into direct contact with the solder paste. A disadvantage of VPS may be rapid heating, while the heat transfer coefficient of vapour is ten times higher than forced convection, this parameter can cause soldering failures such as tombstoning [9]; popcorn cracking [10] (also specified for given package types, such as TQFPs [11]), solder void generation [12]; solder beading [13], and delamination [14] of packages; and solder cracking [15].

Basically, linear soldering temperature profiles can be achieved with VPS, where only the heating gradient can be adjusted via the setting of heating power [16]. However, this linearity does not always match the requirements of solder pastes. Therefore, several modifications were applied on the process such as infrared pre-heating [10] where spluttering of the paste, tombstoning, and damage of the components can be avoided. Also, the dwelling time in the vapour can be reduced for the immersed board. Duck and Zabel of Asscon presented the use of additional cooling around the immersed board to modify the linearity of the profile [16]. They also investigated the application of the vacuum chamber for VPS, where the voids are drawn out from the molten solder [17]. Leicht and Thumm of IBL have investigated the application of the non-saturated vapour space to get non-linear soldering profile [3]. In the non-saturated vapour space the temperature and the concentration of the vapour have a gradient along the Z axis which can be utilized for controlled dipping of the assembled board [3].

Besides the technical innovations mentioned above, there is a very limited amount of examples in the literature for the characterization of vapour space in the soldering tank by measurement or simulation. Therefore, this paper tries to approach VPS from a new perspective and it presents a novel multi-physics model of the vapour phase soldering whereby the vapour concentration and

temperature distribution of the vapour space inside the soldering tank can be studied.

2. Description of the VPS system

The investigated batch type VPS system is based on a closed stainless steel tank. An immersion resistor heater heats the Galden fluid at the bottom of the tank. The power of the heater can be adjusted with a toroidal transformer. The current flow and voltage drop of the heater are measured on-the-fly with a digital measurement setup including an ammeter and a multimeter device. The filament inside the immersion heater has a resistance of 25 Ω . A cooling copper tube is positioned on the top of the tank with circulated ambient temperature water inside it. The block diagram of the system [18] is shown on Fig. 1.

The schematic of the VPS tank is presented in Fig. 2. The dimensions of the tank are the following: width (x axis) 180 mm; length (y axis) 280 mm; height (z axis) 170 mm; and wall thickness 0.5 mm. The metal lid of the tank is removable, with a heat-resistant glass window on the top. The glass window can be removed for in situ access of the vapour space. There is an aperture on the metal lid to access the vapour space with probes and wires.

The horizontal plane of the heater element is immersed in the Galden liquid 10 mm from the bottom of the tank. The pipes of the heater are led out vertically via a prepared outlet at the edge of the

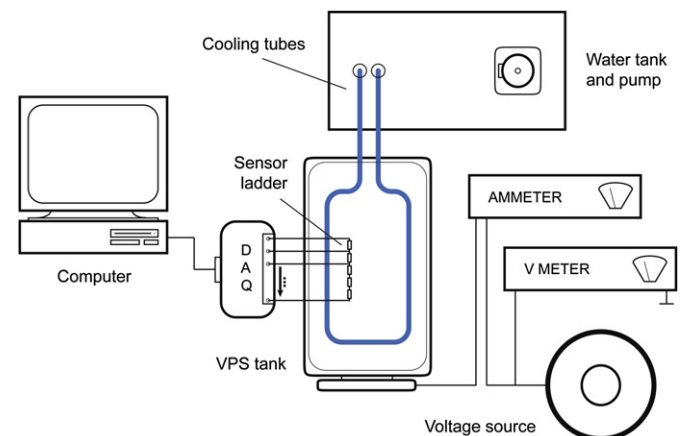


Fig. 1. Block diagram of the VPS system (top view on the VPS tank).

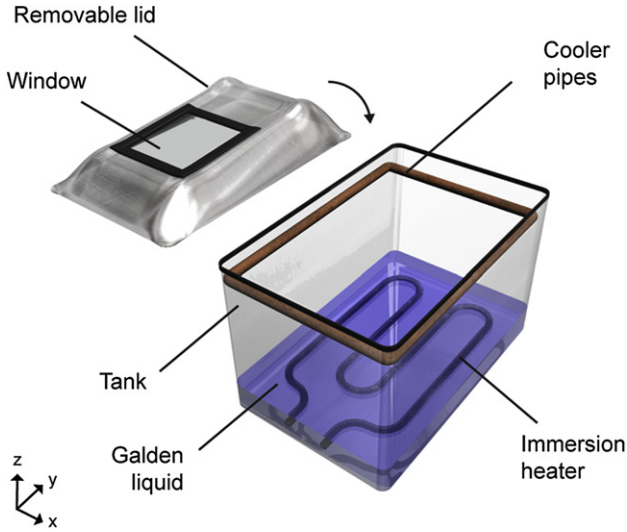


Fig. 2. The VPS tank with the removed lid.

removable lid. However, actual heating only takes place on the horizontal, immersed part of the heater, while the heating filament is only present in that position. The cooler pipe is aligned 1 cm under the top edge of the tank. The cooler pipe is not centred on the y axis; it is aligned to the front wall, due to construction optimizations. The cooling pipes are also led out from the tank via an outlet on the removable lid. The outlets of the heating and cooling systems are omitted from the model and they are also omitted from the schematic image (Fig. 2), which conforms to the model constructed for simulation.

3. Multi-physics model of the VPS system

The multi-physics model of the VPS system is based on the following processes: the heat generated by the immersion heater and the effect of the cooling pipe change the thermal distribution of the oven which can be described by the heat equation [19]:

$$\frac{\partial T}{\partial t} = \frac{\lambda}{\rho \cdot C_s} \cdot \nabla^2 T \quad (2)$$

where t is the time [s], T is the temperature [K], C_s is the specific heat capacity of the material [J/kgK], ρ is the density of the material [kg/m³] and λ is the specific thermal conductivity of the material [W/m K]. If the Galden fluid reaches the boiling point then all influent heat is used for the evaporation [20]:

$$\frac{\partial m}{\partial t} = \frac{\lambda \cdot A}{h} \cdot \nabla T \quad (3)$$

where h is the latent heat of the Galden [J/kg], A is the area crossed by the heat flow [m²] and m is the mass of the phase changing material [kg]. The same phenomenon happens when the Galden vapour condenses on a colder surface. The energy of the phase change generates a heat flow into the solid material.

Concentration difference triggered by the phase changes and inhomogeneous heating of the Galden liquid generates mass transport mechanisms in the fluid itself which can be described by the Fick's law [21]:

$$\frac{\partial \varphi}{\partial t} = D \cdot \nabla^2 \varphi \quad (4)$$

where φ is the concentration of the fluid [kg/m³], D is the diffusion constant of the fluid [m²/s]. Finally the mass transport mechanisms cause energy transport:

$$\frac{\partial E}{\partial t} = C_s \cdot T \cdot \left(\frac{\partial m}{\partial t} + V \cdot \frac{\partial \varphi}{\partial t} \right) \quad (5)$$

where V is the volume wherein the concentration change happens [m³].

3.1. Numerical description

The model was implemented using MATLAB 7.0 software. The model is created with finite difference method (FDM) in order to achieve high solving calculation speed and flexible implementation. The previously defined first and second order Partial Differential Equations (PDEs) can be solved by the three dimensional explicit Forward Time Central Space (FTCS) method [22], in the case of Eqs. (2) and (4):

$$\begin{aligned} v_n(t + \Delta t) = & (1 - 2[r_x + r_y + r_z]) \cdot v_n(t) + r_x \cdot v_{n-1(x)}(t) \\ & + r_x \cdot v_{n+1(x)}(t) + r_y \cdot v_{n-1(y)}(t) + r_y \cdot v_{n+1(y)}(t) \\ & + r_z \cdot v_{n-1(z)}(t) + r_z \cdot v_{n+1(z)}(t) \end{aligned} \quad (6.1)$$

where,

$$r_x = \alpha_x \frac{\Delta t}{\Delta x^2}, r_y = \alpha_y \frac{\Delta t}{\Delta y^2}, r_z = \alpha_z \frac{\Delta t}{\Delta z^2}; \quad (6.2)$$

and in the case of Eq. (3):

$$\begin{aligned} w_n(t + \Delta t) = & w_n(t) - 2[r_x + r_y + r_z] \cdot w_n(t) + r_x \cdot v_{n-1(x)}(t) \\ & + r_x \cdot v_{n+1(x)}(t) + r_y \cdot v_{n-1(y)}(t) + r_y \cdot v_{n+1(y)}(t) \\ & + r_z \cdot v_{n-1(z)}(t) + r_z \cdot v_{n+1(z)}(t) \end{aligned} \quad (6.3)$$

where,

$$r_x = \alpha_x \frac{\Delta t}{\Delta x}, r_y = \alpha_y \frac{\Delta t}{\Delta y}, r_z = \alpha_z \frac{\Delta t}{\Delta z}; \quad (6.4)$$

v and w are general variables where v could be the temperature the concentration and w is the mass, n is the indexing of the mesh, α is a constant parameter, Δt is the time step [s] and $\Delta x, \Delta y, \Delta z$ is the mesh size [m].

The thermal (central) node theory [23] was applied for the implementation of the heat transport between the cells. All cells (Fig. 3) contain a thermal node in their geometrical centre, which represents the thermal mass of the cell as a heat capacity [J/K]:

$$C = C_s \cdot \rho \cdot V \quad (7)$$

The thermal conduction ability of the cells is described with thermal resistances [K/W] between the node and its cell borders towards the different directions:

$$R^x = \frac{\Delta x}{2 \cdot \lambda \cdot \Delta y \cdot \Delta z} \text{ and } R^y = \frac{\Delta y}{2 \cdot \lambda \cdot \Delta x \cdot \Delta z} \text{ and } R^z = \frac{\Delta z}{2 \cdot \lambda \cdot \Delta x \cdot \Delta y} \quad (8)$$

With the thermal node theory applied, the heat equation (Eq. (2)) has the following form:

$$\Delta T_n = \sum_{i=\pm 1(x,y,z)} \frac{T_n - T_{n+i}}{(R_n + R_{n+i}) \cdot C_n} \cdot \Delta t \quad (9)$$

where $i = \pm 1(x,y,z)$ steps mean ± 1 step towards x, y , and z directions (altogether 6 steps).

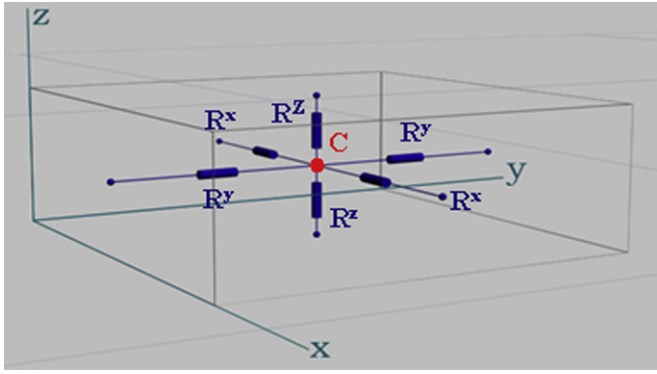


Fig. 3. Numerical grid of the tank with highlighted processing volume.

The most important mass transport in the model is caused by phase changes of the Galden, namely the evaporation of Galden fluid from the tank and the condensation of Galden vapour on colder surfaces in the soldering space and cooling at the top of the tank. With application of the thermal node theory the mass of the phase changed (pc) Galden (Eq. (3)) in a dedicated cell:

$$\Delta m_n^{pc} = \sum_{i=\pm 1(x,y,z)} \frac{T_n - T_{n+i}}{h \cdot (R_n + R_{n+i})} \cdot \Delta t \quad (10)$$

In the case of evaporation the necessary condition of Eq. (10) is that the influent heat has to be positive, which means $T_n < T_{n+i}$. Evaporation causes a mass loss in cell n and results in a mass increase in the cell to where the evaporation happens (these cells are located in the first vapour layer over the Galden liquid). If the evaporated Galden vapour meets a colder surface in the tank it will condensate on the surface. In the case of condensation the necessary condition of Eq. (10) is $T_n = T_{bp} > T_{n+i}$ (where T_{bp} is the boiling point of the Galden). The condensation causes a loss of mass in cell n and results in a mass increase in the cell which cell is located under the condensation area.

The other type of mass transport in the model is caused by diffusion between the fluid cells. If Galden fluid evaporates, the vapour concentration will rise in the cells to where the vapour is transported. The concentration difference between the vapour cells starts the diffusion (diff) of the Galden vapour (vp) in the tank. This type of mass transport can be defined by Fick's law (Eq. (4)):

$$\Delta m_n^{diff} = \sum_{i=\pm 1(x,y,z)} -A_i \cdot D_{vp} \cdot (\varphi_n - \varphi_{n+i}) \cdot \Delta t \quad (11)$$

A_i is the related cross-sectional surface [m^2]. (The diffusion of air caused by the inhomogeneous heating is omitted in the model.)

If a given vapour cell reaches the saturation (sat) concentration ($\sim 20 \text{ kg/m}^3$) it cannot take more Galden vapour. This effect results in a saturation mass transport which is a relatively rapid mass transport compared to the diffusion. The nature of the heating results in the formation of concentration gradients in the x and y directions also, and not only in the z direction, which effect causes different saturation mass transport rates towards the different directions. The phenomenon was implemented in the model so that a saturated vapour cell distributes the received amount of vapour beyond saturation between the adjacent vapour cells in different ratio:

$$\Delta m_n^{st} = -(m_n(t) - m_{\max}) + k_{xy} \cdot \sum_{i=\pm 1(x,y)} (m_{n+i}(t) - m_{\max}) + k_z \cdot (m_{n-1(z)}(t) - m_{\max}) \Big|_{m > m_{\max}} \quad (12)$$

where $\pm 1(x)$, $\pm 1(y)$ and $-1(z)$ means ± 1 and -1 step on the x , y and z axis. The ratios (k_{xy} and k_z) were determined by iteration of the model (see in the verification section).

During the verification of Galden heating (see in Section 4.) it was observed that some kind of additional energy transport mechanism – beyond the heat conduction – has to be present between the Galden liquid cells to reach the relatively fast temperature response to the heating. This additional energy transport is also caused by mass diffusion between the Galden liquid cells. The inhomogeneous heating of the Galden liquid causes a big temperature gradient and a locally non-negligible thermal expansion of the Galden liquid which effect starts mass transport between the cells.

The thermal expansion of the Galden cell can be defined:

$$\Delta V_n = \beta \cdot V_n(t) \cdot (T_n(t) - T_n(t + \Delta t)) \quad (13)$$

where β is the thermal volumetric expansion coefficient of the Galden liquid [$m^3/m^3 \text{ K}$]. In fixed model geometry the thermal expansion (txp) of the liquid in a dedicated cell can be described as an amount of mass which leaves the cell. This causes mass change between the adjacent cells as a “blending” of the liquid. It is supposed that the expansion is even across the different directions; therefore in a given direction it depends on the surface ratio of the cell border (where the mass leaves) and the total surface of the cell:

$$\Delta m_n^{txp} = - \sum_{i=\pm 1(x,y,z)} \left(\rho_n \cdot \Delta V_n \cdot \frac{A_i}{A_n} - \rho_{n+i} \cdot \Delta V_{n+i} \cdot \frac{A_i}{A_{n+i}} \right) \quad (14)$$

where ρ is the density of the Galden liquid [kg/m^3], A_i are the area of cell borders [m^2] and A_n is the total surface of the cell [m^2]. The forming density difference – caused by the different thermal expansion of the cells – initializes mass diffusion between the Galden liquid cells also, ($\Delta m_n^{diff(lig)}$) which can be described with Eq. (11).

Evaporation, condensation and expansion of the Galden fluid will cause a minor change in the position of the vapour–liquid interface in the pool. According to our calculations, the evaporation and condensation would cause a maximum of -0.7 mm change, and the expansion would cause a maximum of $+0.9 \text{ mm}$ change if they do not compensate each other. (The calculation was performed during a 20 min operation of the oven). Due to the compensation, our approximation shows a change of the vapour–liquid interface between 0 and 0.3 mm. This value can be neglected due to the two magnitude difference in the order of discretization. (The upper limit of the approximation was extended from 0.2 to 0.3 mm since the expansion of the Galden starts earlier than the evaporation.)

The mass transport and the mass change induced energy transport balance equations for a vapour cell according to Eq. (5), (10)–(12) and (14) are the following:

$$\Delta m_n^{vp} = \sum_{i=1}^j \Delta m_{n+i}^{ev} - \Delta m_n^{co} + \Delta m_n^{diff(vp)} + \Delta m_n^{sat} \quad (15)$$

$$\Delta E_n^{vp} = C_{ga} \cdot T_{bp} \cdot \Delta m_n^{vp} \quad (16)$$

where j is the number of the cells which evaporate into the dedicated cell. The mass transport and the mass change induced energy transport balance equations for a liquid cell according to Eqs. (5), (10), (11), (14) and (15) are the following:

$$\Delta m_n^{liq} = -\Delta m_n^{ev} + \Delta m_n^{txp} + \Delta m_n^{diff(liq)} \quad (17)$$

$$\Delta E_n^{\text{liq}} = -h \cdot \Delta m_n^{\text{ev}} + C_{\text{ga}} \cdot \sum_{i=\pm 1(x,y,z)} \left(-T_n \cdot \left[\rho_n \cdot \frac{A_i}{A_n} \cdot \frac{dV_n}{dt} - A_i \cdot D_{\text{ga}} \cdot (\rho_n - \rho_{n+i}) \right]_{\rho_n > \rho_{n+i}} + T_{n+i} \cdot \left[\rho_{n+i} \cdot \frac{A_i}{A_{n+i}} \cdot \frac{dV_{n+i}}{dt} + A_i \cdot D_{\text{ga}} \cdot (\rho_n - \rho_{n+i}) \right]_{\rho_n < \rho_{n+i}} \right) \quad (18)$$

where the inflowing and outflowing mass has to be distinguished. The energy and mass transport caused by the back flow of the condensed Galden liquid into the pool is also omitted. (Most of the condensed Galden remains on the surface where the condensation happens). The condensation results in energy gain of the solid (sol) cells on which the condensation happens:

$$\Delta E_n^{\text{sol}} = h \cdot \sum_{i=\pm 1(x,y,z)} \Delta m_{n+i}^{\text{co}} \quad (19)$$

The temperature balance equation of the model (valid for all types of cells) can be defined according to the heat conduction (Eq. (9)) and the mass change induced energy transports (Eqs. (16), (18) and (19)):

$$\Delta T_n = \sum_{i=\pm 1(x,y,z)} \frac{T_n - T_{n+i}}{C_n \cdot (R_n + R_{n+i})} \cdot \Delta t + \frac{\Delta E_n}{C_n} \quad (20)$$

The following boundary conditions were applied: the initial temperature of the cells was:

$$T_n(0) = 25^\circ \text{C} \quad (21.1)$$

The heating of the Galden liquid and the cooling of the vapour space in the tank was implemented as a fixed temperature heating/cooling with constant temperatures and infinite heating/cooling capacity.

$$T_n(t)|_{n=\text{heating}} = 238^\circ \text{C}; \quad T_n(t)|_{n=\text{cooling}} = 25^\circ \text{C} \quad (21.2)$$

According to preliminary measurements, these circumstances approximate reality. By applying this heating parameter, the maximum input power density was 5 W/cm² which is the most suitable for the Galden fluid according to the material parameters. If the input power is too high, the polymer chains of the Galden fluid would decompose before evaporation and decrease the efficiency of the evaporation. Heat and mass transfer are omitted across the solid boundaries of the model (the walls of the VPS tank):

$$\lambda \cdot \nabla T_n|_{n=\text{out}} = 0 \quad (21.3)$$

$$D \cdot \nabla^2 \varphi_n|_{n=\text{out}} = 0 \quad (21.4)$$

3.2. Geometry model and space discretization

The geometry model contains cuboid and cubic cells in a non-uniform grid. Equilibrium is assumed in all cells; therefore, during cell partition, borders of the different materials have to be taken into consideration. Our model contains the following types of cells: stainless steel (the wall of the tank and the lid), Galden liquid, copper (the wall of the cooler tube), vapour cells (mixture of air and Galden vapour) and the heater tube as a composite material (Wolfram filament in ceramics lining with stainless steel tube). Most of the physical properties of the cells can be found in the literature, but the properties of the composite cells were calculated according to the mass ratio and size ratio of the different composing materials in the cell (the method was presented in [23]).

The applied physical parameters of the materials can be seen in Table 1.

The construction of the grid was achieved via section planes where the cell borders are the section lines of these planes. This method ensures that all neighbouring faces of the cells should have the same size and all cells can have a maximum of 6 neighbours. Due to the mesh size influence in obtaining the results, the values of Δx , Δy and Δz were investigated by reducing them from a reference value to a specific value from which numerical solution shows no significant variation. An acceptable resolution was found where the values of Δx , Δy and Δz were between 0.5 and 10 mm (depending on the material borders) with 20 cells along the x (shorter side of the tank), with 30 cells along the y axis (longer side of the tank) and with 18 cells along the z axis (height of the tank). In this case, the numerical grid contains 11,800 cells. The applied grid can be seen in Fig. 4. In order to allow a better visualization of this concept, only the section lines of the base planes are visible in the picture. The processing volume of the tank (where the soldering takes place) was defined above the heater element, between 20 and 160 mm along the x axis; 40–250 mm along the y axis and 40–160 mm along the z axis (marked by the purple column in Fig. 4).

The stability and the convergence criterion of Eq. (6) according to Li et al. [24] is:

$$r_x + r_y + r_z \leq 0.5 \quad (22)$$

During the stability and convergence analysis of the model, the previously introduced space resolution was used and the maximum time step was aimed for the fixed space resolution. It was found that the model converges and is stable if $\Delta t < 0.26$ s. The Truncation Error (TE) of the applied FTCE method is $O(\Delta t, (\Delta x)^2, (\Delta y)^2, (\Delta z)^2)$. In order to reduce TE without the increase of the model complexity only the time step was reduced to $\Delta t = 0.1$ s. However, this increased the number of calculation steps, increasing the round error. Therefore double words were applied for all variables. The gratuitous memory demand was avoided by the application of “sparse vectors”, which means that memory space is reserved only for those elements of the vectors where the stored value is not zero. In addition, the application of sparse vectors increases the

Table 1
Physical parameters of applied materials.

	Density [kg/m]	Specific heat capacity [J/kg K]	Specific thermal cond. [W/m K]	Latent heat [J/kg]	Diffusion coeff. [cm ² /s]	Thermal exp. coeff. [m ³ /m ³ K]
Air	1	1009	0.03	—	—	—
Galden liquid	1820	973	0.77	63,000	10 ^{−9b}	0.1
Galden vapour	20 ^a	973	0.07	63,000	0.35 ^b	—
Stainless steel	8000	500	18	—	—	—
Composite heater tube	4945	649	22.96	—	—	—
Cooper	8960	390	401	—	—	—

^a Saturation vapour concentration at 180 °C.

^b At 180 °C.

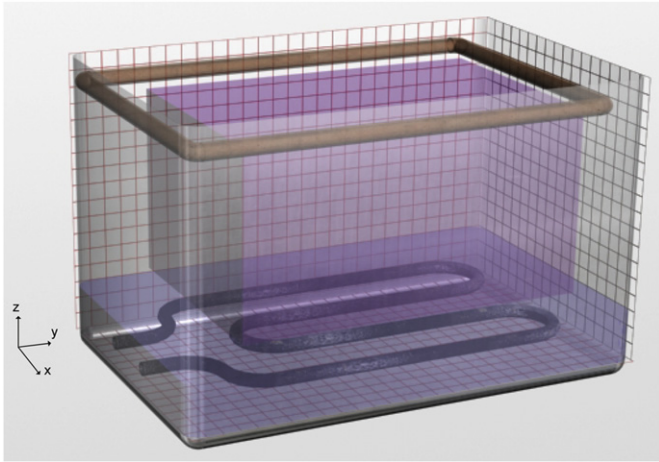


Fig. 4. The basic thermal cell.

calculation velocity because only those calculation steps are done where the sparse vectors have elements different from zero.

3.3. Verification of the model

The model was verified via dynamic temperature and pressure measurements in the vapour space of the tank. Although the model calculates the concentration of the Galden vapour, the direct concentration measurements would be too complicated; therefore, this part was verified by pressure measurements and with the comparison of the differential concentration change and differential pressure change characteristics.

For the temperature and pressure measurements, an optional sensor ladder was developed. The sensors are affixed to a rod vertically at custom heights and then the rod is positioned in the tank perpendicular to the bottom. Heat-resistant, flexible silicone rubber hoses were also affixed to this rod for the pressure measurements. The temperature measurements were carried out with K-type thermocouples (absolute measurement accuracy was $\pm 0.5^\circ\text{C}$) and the pressure measurements were carried out with Sensirion SDP1108 precision differential manometers (the measurement accuracy in the applied measurement range was $\pm 0.002\text{ Pa}$).

The wires of the thermocouples have Teflon coating to withstand the high temperatures inside. Silicon rubber hoses with a diameter of 5 mm were applied to the SDP1108 sensor ports to transfer the pressure changes into the manometers. The altitude and the hose dimensions were compensated on the final measurement data. The analogue values are recorded with a Novus MyPCLab USB data logger device. The measurements were carried out at three different locations which were located diagonally in the tank in the following positions (x,y) [mm]: (30,30); (100,150) and (180,270). Three different measurement heights were applied at all locations in the following positions (z) [mm]: (50), (100) and (150). The measurements were repeated three times.

In Fig. 5 the calculated and measured temperature values are compared at the base location of (x,y) [mm]: (100,150) on the three different height (z) levels mentioned above. The results show satisfying conformity, the average relative deviation between the calculated and measured values was 1.71%. However, minor deviations were observed between the measured and calculated values mainly in the case of the 100 and 150 heights after the breaking points of the curves. The increase of the measured values is a bit slower than the increase of the calculated values. The reason could

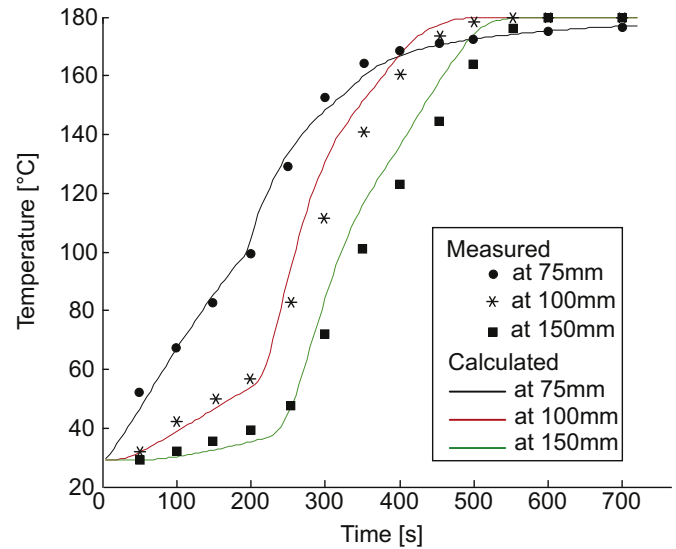


Fig. 5. Calculated and measured temperature values in the vapour space (x,y):(100,150).

be that the adiabatic condition is not an entirely accurate approach between the tank and the environment.

The developing Galden vapour will form a concentration gradient in the tank, which results in a pressure gradient also. After the temperature saturation reaches a given height, the space below can be considered nearly isothermal (due to the very low thermal conductivity of the Galden vapour). Therefore, the change in the concentration gradient – caused by the diffusion or the condensation – in the given volume will have to be followed by the pressure gradient change with the same characteristics, according to the general gas law. In Fig. 6 the change of the calculated concentration gradients are compared to the change of the measured pressure gradients between the three different height levels and the top of the tank (z) [mm]: (170) at the location (x,y) [mm]: (100,150). The temperature saturation points – when the saturation is more than 90% – are also marked in Fig. 6 (ST1, ST2 and ST3) according to Fig. 5. Fig. 6 reveals that the measured and simulated characteristics conformed very well.

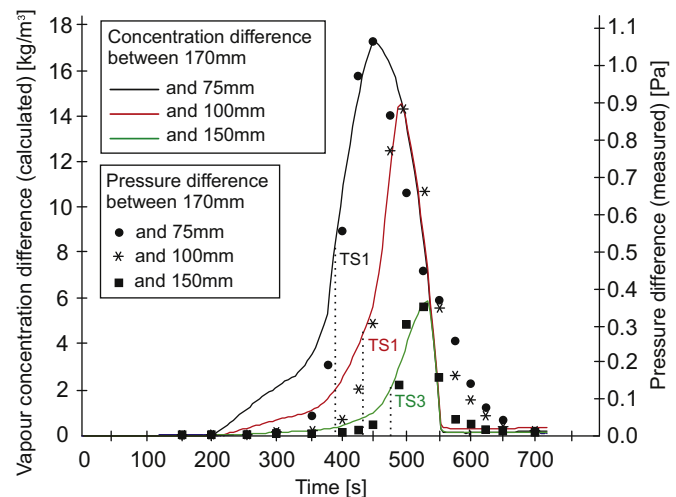


Fig. 6. Calculated concentration and measured pressure differences in the vapour space (x,y):(100,150).

According to the calculations, the best match was received when the following saturation mass transport ratios (Eq. (8)) were used: $k_{xy} = 0.115$ valid to the $\pm x$ and $\pm y$ directions and $k_z = 0.54$ valid to the $-z$ direction (it was assumed that the back flow from the $+z$ direction is impossible). The delay transients of the pressure difference vary mainly at the decreasing phase, caused by the construction of the pressure sensor, which needs a definite minimum gas flow for measuring, due to the dynamic measurement principle.

4. Results and discussion

The main aim of the simulations was to investigate the previously defined processing volume of the tank and whether it is suitable for soldering, or if this custom defined volume has to be modified. As previously discussed, the key factors in vapour phase soldering are the temperature and vapour concentration in the vapour phase. The most effective soldering can be achieved if the vapour temperature and concentration is already saturated in the processing volume before the soldering. (Discussing non-saturated vapour is an aim of a future work.)

Therefore in the first step, temperature and concentration distributions of the vapour space were studied along different section planes and at different times in order to find the inhomogeneities of the system caused by the immersion heating. Afterwards, the saturation heights of the temperature and vapour concentration were calculated in the vapour space in order to study the fill of the processing volume of the tank by the vapour and to define the necessary heating time for the saturation. As discussed above, the vapour concentration saturation is 20 kg/m^3 (defined by the Galden manufacturer) and the temperature saturation is 180°C as this is the boiling point of the applied Galden.

In Fig. 7 the temperature and vapour concentration distributions can be seen along the plan (y) [mm]:150 at different time points calculated from the start of the heating. The top of the Galden liquid is marked by the dark dashed line and the previously defined borders of the processing volume are marked by the white dashed rectangles.

It was observed during the simulations (along all y section planes of the tank) that the saturation of the vapour concentration

began above the two close inner tubes of the heating element (Fig. 7a, b) while the saturation of the temperature in the vapour space started between the middle tubes (Fig. 7d, e). It was expected that the temperature of the vapour space will saturate together with the vapour concentration, thus the mass of the Galden molecules is around twenty-five times more as the mass of the air molecules.

The effect can be explained with the following: the evaporation is the most active above the closer tubes due to the highest temperature values of the Galden is in this volume; while the temperature saturation of the vapour space can be more successful in the middle of the tank due to the cooling effect of the tank walls. However, it does not cause any irregularity during the soldering process, since the effect disappeared after a 12 min heating of the system (Fig. 7c, f), and the defined processing volume is suitable along the y axis. It was also found that the cooling is sufficient along the closing line of the roof of the tank, minimizing the loss of Galden (Fig. 7c).

Since the heater system is asymmetric along the y direction (the heater element is closer to the right wall as the left wall, Fig. 7), some inhomogeneities were expected in the vapour space at the x section plains. This is most visible along the x median of the tank where there is only an x-parallel connection tube between the y-parallel heating tubes.

In Fig. 8 the temperature and vapour concentration distributions can be seen along the plane (x) [mm]:100 at different time points calculated from the start of the heating. The top of the Galden liquid is marked by the dark dashed line, and the previously defined border of the processing volume is marked by the white dashed rectangles.

As it was expected, the inhomogeneity of the vapour space is considerable at the beginning of the heating (Fig. 8a, c), and some inhomogeneity also remains after 12 min of heating, like in the case of vapour concentration distribution (Fig. 8b). It has to be noted that the “worst case scenario” was presented here and this effect is much weaker at the other x section planes. However, our assumption that the best position of the processing volume is over the heater elements seemed to be wrong. Despite the asymmetrical position of the heater element in the tank, the most effective saturation of the vapour space did not happen above the heater

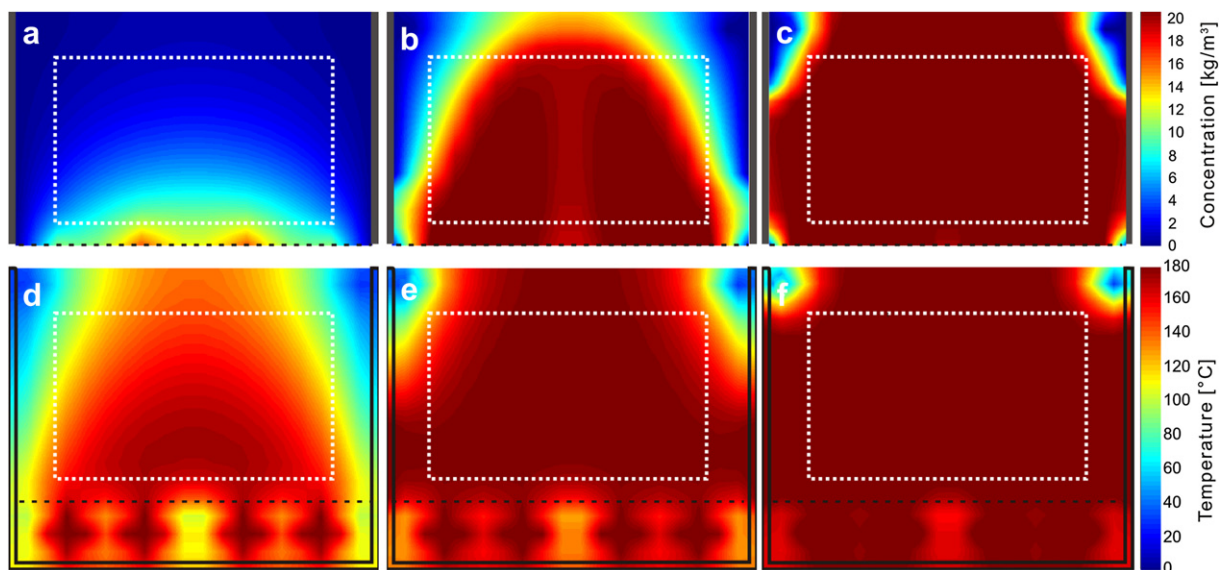


Fig. 7. (a–c) Vapour concentration and (d–f) temperature distributions along (y) [mm]:150 at 7, 9 and 12 min.

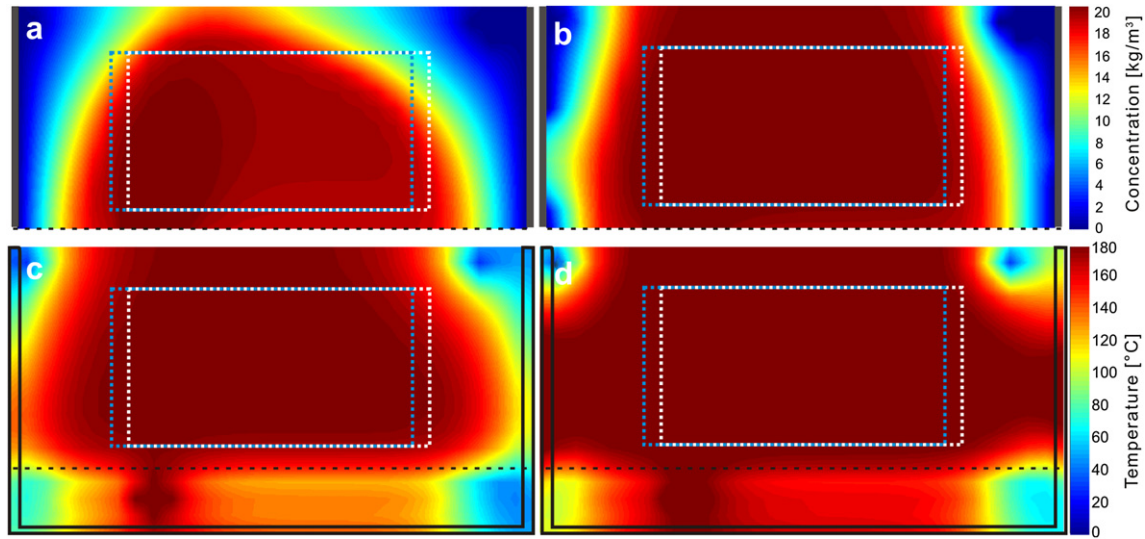


Fig. 8. (a–b) Vapour concentration and (c–d) temperature distributions along (x) [mm]:100 at 7 and 12 min.

element, but slightly to the left of it along the y axis. Therefore, the processing volume has to be moved along the y axis towards the left by 10 mm (marked by the blue dotted rectangles in Fig. 8).

In Fig. 9 the saturation heights of the vapour concentration and the temperature can be seen in the vapour space at different times calculated from the start of the heating. In this case the different colours mark the highest points in the vapour space of the tank where the saturation is already taking place. The previously defined border of the processing volume is marked by the white dashed rectangles and the newly defined marked by the blue dashed rectangles.

The different behaviour of the temperature and the vapour concentration saturation is clearly visible here as well (Fig. 9a and d) – where concentration saturates above the two closer heater tubes, while temperature saturates between the middle heater tubes. As it was expected according to the previous results (Figs. 7

and 8), the saturation already took place in the processing volume after 12 min heating. In addition, it is also shown here that the processing volume has to be moved to the left along the y axis by 10 mm. However some inhomogeneities still exist in the saturated vapour concentration (Fig. 9c) at the left and right borders of processing volume but it is located over the 160 mm vertical mark, just over the processing volume.

It has to be noted that immersion heating tubes cause considerable inhomogeneities in the evaporation of the Galden liquid. The boiling of the Galden and the most of the evaporation happens only in the close environment of the heater tubes (Figs. 7 and 8). However this effect does not cause major problem during the filling of the processing volume by the vapour. According to the simulation results with a minor change of the previously defined processing volume the system is suitable for soldering after 12 min of heating with the given heating power.

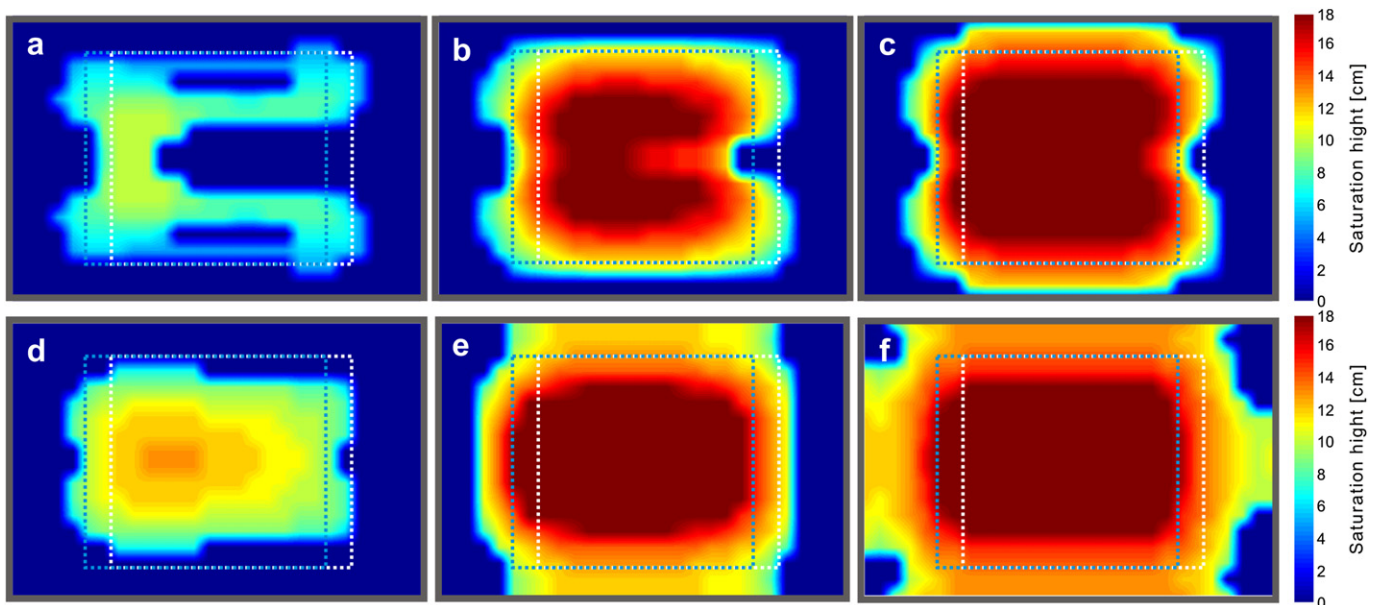


Fig. 9. Saturation heights of (a–c) the vapour concentration (20 kg/m^3) and (d–f) the temperature (180°C) in the tank at 8, 10 and 12 min.

5. Conclusions

A multi-physics model of the vapour phase soldering process was developed. The model takes into consideration the following transport mechanisms: heat transport by heat conduction; mass transport by diffusion between the fluid cells and by phase change during the evaporation and condensation of the Galden liquid, as well as energy transport caused by the mass transports. The model capability was tested by simulations of a batch type vapour phase soldering system in order to find the optimal settings.

According to the simulated temperatures, concentration distributions and saturation heights of the vapour space, the main conclusions are the following: the applied heating (heater tubes) causes considerable inhomogeneities in the evaporation of the Galden liquid. The boiling of the Galden and most of the evaporation happens only in the close environment of the heater tubes (Figs. 7 and 8). Therefore, the saturation of the vapour concentration starts above heater tubes (Fig. 7a, b). Contrarily, the saturation of the temperature cannot follow the concentration – it starts in the middle of the tank (Fig. 7d, e) due to the cooling effect of the tank walls. However, this effect does not cause major problem during the filling of the processing volume by the vapour. Our assumption that the best position of the processing volume is over the heater elements needed correction. The asymmetrical position of the heater element along the y axis in the tank resulted in the finding, that the most effective saturation of the vapour space did not happen above the heater element, but to the left of it along the y axis. Therefore, the processing volume has to be defined above the heater elements but shifted 10 mm to the left along the y axis. According to the simulation results, the studied VPS system is ready for soldering after 12 min of heating (with the given parameters).

Further work is required to test the limits of our model by real soldering simulations with board samples immersed into the processing volume, and to reveal how to handle the lowering and elevation of the soldered circuit into and from the tank effectively. In addition, the modelling approach presented in this paper is also applicable to simulation and optimization of other thermal processes where phase change takes place.

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